

# ESE 6510 Note 2: Variance Reduction

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## 1 Variance in Policy Gradient Estimation

Although the policy gradient

$$\mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[ \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) R(\tau) \right]$$

is unbiased, it typically suffers from high variance. Since policy updates are performed using stochastic gradient ascent, noisy gradients can lead to unstable learning and slow convergence.

One major source of variance is that early policy gradient terms are multiplied by the total return of the entire trajectory. In long-horizon tasks, stochastic effects far in the future can significantly alter the total return, causing large fluctuations in the gradient contributions of early actions.

To make this dependence explicit, consider the gradient of the expected reward at a fixed timestep  $t$ :

$$\nabla_{\theta} \mathbb{E}_{\tau \sim p_{\theta}(\tau)} [r_t].$$

Applying the score function identity yields (using  $T - 1$  since no reward is received after the terminal transition)

$$\nabla_{\theta} \mathbb{E}[r_t] = \mathbb{E} \left[ \sum_{j=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_j \mid s_j) r_t \right].$$

However, the reward  $r_t$  cannot depend on actions taken after time  $t$ . As a result, the corresponding terms contribute zero in expectation, and the expression above describes the contribution of a *single timestep*  $t$ :

$$\nabla_{\theta} \mathbb{E}[r_t] = \mathbb{E} \left[ \sum_{j=0}^t \nabla_{\theta} \log \pi_{\theta}(a_j \mid s_j) r_t \right].$$

We now consider the entire trajectory by summing the contributions of all rewards  $r_0, \dots, r_{T-1}$ :

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} \mathbb{E} \left[ \sum_{t=0}^{T-1} r_t \right] = \mathbb{E} \left[ \sum_{t=0}^{T-1} \sum_{j=0}^t \nabla_{\theta} \log \pi_{\theta}(a_j \mid s_j) r_t \right].$$

Reordering the summations yields

$$\nabla_{\theta} J(\theta) = \mathbb{E} \left[ \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \underbrace{\sum_{j=t}^{T-1} r_j}_{\text{reward-to-go}} \right].$$

This form shows that each policy gradient term is weighted only by rewards that occur at or after the corresponding timestep. Removing dependence on unrelated future actions preserves the expected gradient while reducing unnecessary variance.

The reward-to-go construction is one example of modifying the return signal to reduce variance. More generally, we may ask whether other transformations of the return can reduce variance without changing the expected policy gradient.

## 1.1 Why Baselines Do Not Change the Expected Policy Gradient

From the previous section, we ended with the reward-to-go policy gradient:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[ \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \underbrace{\sum_{j=t}^{T-1} r_j}_{\text{reward-to-go}} \right].$$

Now consider subtracting a state-dependent baseline inside the same bracket:

$$\nabla_{\theta} J_b(\theta) = \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[ \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left( \sum_{j=t}^{T-1} r_j - b(s_t) \right) \right]. \quad (1)$$

Expanding the product inside the sum gives

$$\sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \sum_{j=t}^{T-1} r_j - \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) b(s_t).$$

Using linearity of expectation,

$$\mathbb{E}[X - Y] = \mathbb{E}[X] - \mathbb{E}[Y], \quad \mathbb{E} \left[ \sum_t Z_t \right] = \sum_t \mathbb{E}[Z_t],$$

we obtain

$$\nabla_{\theta} J_b(\theta) = \nabla_{\theta} J(\theta) - \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[ \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) b(s_t) \right].$$

Thus, subtracting a baseline does not change the expected gradient provided that

$$\mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[ \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) b(s_t) \right] = 0.$$

It therefore suffices to show that for each timestep  $t$ ,

$$\mathbb{E}_{\tau \sim p_{\theta}(\tau)} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) b(s_t)] = 0.$$

We now split the trajectory into its prefix and suffix using the law of total expectation:

$$\mathbb{E}_{\tau} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) b(s_t)] = \mathbb{E}_{s_{0:t}, a_{0:t-1}} [\mathbb{E}_{s_{t+1:T}, a_{t:T} | s_{0:t}, a_{0:t-1}} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) b(s_t)]] .$$

The score term depends only on  $(s_t, a_t)$ , and the baseline depends only on  $s_t$ . Conditioned on the prefix  $(s_{0:t}, a_{0:t-1})$ , the only remaining randomness inside the inner expectation is the action  $a_t \sim \pi_{\theta}(\cdot | s_t)$ . Hence the inner expectation reduces to

$$= \mathbb{E}_{s_{0:t}, a_{0:t-1}} [b(s_t) \mathbb{E}_{a_t \sim \pi_{\theta}(\cdot | s_t)} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)]] .$$

Finally,

$$\mathbb{E}_{a_t \sim \pi_{\theta}(\cdot | s_t)} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)] = \sum_{a_t} \nabla_{\theta} \pi_{\theta}(a_t | s_t) = \nabla_{\theta} \sum_{a_t} \pi_{\theta}(a_t | s_t) = \nabla_{\theta} 1 = 0.$$

Substituting this back yields

$$\mathbb{E}_{\tau} [\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) b(s_t)] = 0,$$

so subtracting a state-dependent baseline does not change the expected policy gradient.

#### Why $b(s_t) = \mathbb{E}_{a_t} [f(s_t, a_t)]$ is still a valid baseline

A common question is whether a baseline of the form

$$b(s_t) = \mathbb{E}_{a_t \sim \pi_{\theta}(\cdot | s_t)} [f(s_t, a_t)]$$

is allowed, since  $f$  depends on the action.

Once the expectation over  $a_t$  is taken,  $b(s_t)$  becomes a deterministic function of  $s_t$  only, and therefore can be treated as constant inside the inner expectation over  $a_t$  in the unbiasedness proof.

In particular,

$$V^{\pi}(s_t) = \mathbb{E}_{a_t \sim \pi_{\theta}(\cdot | s_t)} [Q^{\pi}(s_t, a_t)] ,$$

so  $V^{\pi}(s_t)$  depends only on  $s_t$  and is therefore a valid baseline.

## 1.2 Why Subtracting a Baseline Reduces Variance

Having established that subtracting a state-dependent baseline preserves the expected policy gradient, we now examine how the choice of baseline affects the *variance* of the estimator.

Let's consider a single summand of the policy gradient Eq. 1

$$\nabla_{\theta} J_b^t(\theta) = \mathbb{E}_{(s_t, a_t)} \left[ \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \left( \sum_{j=t}^{T-1} r_j - b(s_t) \right) \right] , \quad (2)$$

and study the dependence between its variance and the baseline. To simplify the notation, we define

$$\ell_t \triangleq \nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t), \quad R_t \triangleq \sum_{j=t}^{T-1} r_j, \quad b_t \triangleq b(s_t).$$

The policy gradient at time  $t$  is then

$$g_t \triangleq \mathbb{E}_{(s_t, a_t)} [l_t(R_t - b_t)]$$

The variance of this single-step gradient is

$$\text{Var}[g_t] = \mathbb{E}[\ell_t^2(R_t - b)^2] - (\mathbb{E}[\ell_t R_t])^2,$$

where we used  $\mathbb{E}[\ell_t b] = 0$  from the previous section.

Expanding the quadratic term gives

$$\mathbb{E}[\ell_t^2 R_t^2] - 2b \mathbb{E}[\ell_t^2 R_t] + b^2 \mathbb{E}[\ell_t^2].$$

Differentiating with respect to  $b$  and setting the derivative to zero yields the variance-minimizing baseline at this timestep:

$$b^* = \frac{\mathbb{E}[\ell_t^2 R_t]}{\mathbb{E}[\ell_t^2]} \approx \mathbb{E}[R_t] = V^{\pi}(s_t) \quad (3)$$

Eq. 3 shows that the optimal baseline is the weighted sum of expected returns. That is not too far from the expected return from the current state to the end of the episode, i.e., the value function. This gives us:

$$\nabla_{\theta} J(\theta) = \mathbb{E} \left[ \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) \left( \sum_{k=t}^{T-1} r_k - V^{\pi}(s_t) \right) \right] \quad (4)$$

We generally call the quantity inside the parentheses *the advantage function*:

$$A^{\pi}(s_t, a_t) = \sum_{k=t}^{T-1} r_k - V^{\pi}(s_t).$$

Note that the derivation above provides intuition for variance reduction, but is not formal; a more formal treatment can be found in Greensmith et al. (2004).